Abu Jafar Saifullah

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Research Interests

AI for Software Engineering, Software Engineering, AI & Machine Learning, Web Accessibility

Education

University of Dhaka — Dhaka, Bangladesh
M.Sc. in Software Engineering (completed; final result pending); ***GPA: 3.92/4.00*** — *Apr 2025 – Sep 2026*
B.Sc. in Software Engineering; ***GPA: 3.89/4.00*** — *Jan 2019 – Dec 2023*

Notre Dame College — Dhaka, Bangladesh
Higher Secondary Certificate; ***GPA: 5.00/5.00*** — *Jun 2016 – Jun 2018*

Publications

Abu Jafar Saifullah, Tasmia Zerin, Zerina Begum, Kazi Sakib. *"NavA11y: A Dynamic Analysis Approach for WCAG 2.4 Focus-Behavior Evaluation."* In *Proceedings of the 21st International Conference on Evaluation of Novel Approaches to Software Engineering* **(ENASE 2026)**, vol. 2, pp. 1019–1030. SCITEPRESS. Benidorm, Spain. *Full paper; presented in person, May 2026.* DOI: 10.5220/0015045700004015 [Paper] [Code]

Method: Drives a real browser through each page's keyboard tab order and, at every focus stop, checks that focus lands where expected, is visibly indicated, and is not hidden behind other content.
Results: 100% precision and recall on a 22-page labeled dataset; 2,947 violations on 26 production websites at 90% precision, validated by two independent expert raters; first tool to cover all five WCAG 2.4 focus-behavior criteria.

Syed Fatiul Huq, Md. Asif Haider, **Abu Jafar Saifullah**, Thomas Zimmermann, Sam Malek. *"Vibing Accessibly: Investigating the Accessibility of Websites Generated by LLM Coding Agents." Under review, 2026.*

Method: Generated 90 websites with three coding agents and four language models, audited complete user journeys with three accessibility checkers, re-ran 24 configurations with agent accessibility Skills, and ran user testing with blind participants.
Results: None of the 90 websites was fully accessible (80 violating elements on average, 13 WCAG criteria violated); Skills significantly improved accessibility scores but did not remove all violations and lowered functional correctness.

Research Experience

DSSE Research Group, IIT, University of Dhaka — Dhaka, Bangladesh
Graduate Researcher; supervisor Dr. Zerina Begum, co-supervisor Dr. Kazi Sakib — *Apr 2025 – Sep 2026*

- M.Sc. thesis (completed): *Detecting and Repairing Keyboard Focus-Behavior Violations through Dynamic Analysis.* Built **NavA11y** for detection and **RepairA11y** for repair.
- Designed NavA11y's two-phase analysis: page level drives real keyboard navigation and compares observed vs. expected focus order; element level checks each focused element is visible, has a focus indicator, and is not hidden.
- Extended the GDS Accessibility Tool Audit's 6 focus test pages with 16 new pages into a 22-page labeled dataset (*Dd*) covering all five WCAG 2.4 focus-behavior SCs, including pass, boundary, and obscuration cases; released for evaluating future tools.

SEMERU Lab, William & Mary — Remote
Research Intern, supervised by Dr. Denys Poshyvanyk — *Aug 2023 – Dec 2023*

- Built a programming language-specific tokenizer to improve NLP model predictions for software engineering tasks, contributing to the lab's research on intelligent software tools.

Industry Experience

Optimizely, Content Marketing Platform — Dhaka, Bangladesh
Software Engineer II (promoted from SE I, Apr 2026) — *Apr 2024 – Present, Full-time*

- Delivered end-to-end integration of an internal AI SDK by moving its service from Express to Hono, adding the AWS CodeArtifact publishing pipeline, and instrumenting AI-context events across **10+ product surfaces**.

- Shipped the platform's block-based AI document UI (Adaptive-Cards-style): backend block-context extraction for LLM prompts, frontend block-form interaction, and AI resource rendering.
- Moved the platform onto the public **Axiom** design system (`@optiaxiom/react`) by leading the migration with **26 merged upstream PRs** and writing the Migration and Icon Contribution Guides other teams adopted; brought Dark Mode to **15+ surfaces**; co-built **Spotlight** search, winner of **Optimizely Global Hackathon 2025**.

Orion Informatics Ltd. Dhaka, Bangladesh

Software Engineer *Jan 2023 – Jan 2024, Full-time*

- Built data visualisation and reporting features for an Attendance Management System; developed a two-service ASP.NET Core webhook relay for TradeStation alerts deployed on Azure.

Research Projects

NavA11y | *Node.js, Playwright, Chromium, GitHub Actions* | Code | *ENASE 2026*

- Dynamic WCAG 2.4 accessibility testing tool, the first to cover all five focus-behavior success criteria. Drives a real browser through each page's keyboard navigation and inspects what is actually rendered at every focus stop. 100% precision/recall on a 22-page labeled benchmark; 2,947 violations on 26 production sites; 90% precision confirmed by independent expert raters; dataset released for community benchmarking.

RepairA11y | *Node.js, Playwright, Vitest* | *M.Sc. Thesis, 2026*

- Repairs WCAG 2.4 focus-behavior violations from the runtime evidence NavA11y records: writes a typed patch (a CSS rule or an attribute change) with deterministic rules or a language model, applies it in a live browser, and re-runs NavA11y's checks. On 32 fixture cases, rules fixed 29 and the model 25, with no new focus-behavior violation; runtime evidence raised the model from 5 to 7 of 7 realistic focus-appearance cases.

Clippy | *Angular, Node.js, AWS S3* | GitHub | *ICSE SCORE 2023, Top 6*

- Smart PDF reader for researchers with knowledge-graph extraction, citation-network visualization, and paper recommendation; Top 6 finalist in the ICSE 2023 Student Contest on Software Engineering (SCORE).

BPE Tokenizer | *Python, Flask, HuggingFace* | PyPI

- Byte Pair Encoding tokenizer implemented from scratch and published as `python-bpe-tokenizer` on PyPI; includes a Flask web interface for training on custom corpora. Built as direct precursor to SEMERU Lab tokenizer research.

ML Bug Mining (SZZ) | *Python, GitPython, pandas, scikit-learn* | *M.Sc. Coursework*

- Mined ML-related bugs across 9 open-source repositories using the SZZ algorithm; identified bug-fixing and bug-introducing commits via keyword heuristics and git blame, filtered for ML-library-touching code.

Other Projects

Blueberry Auto | *React, Hono, Supabase, Vercel* | Live | *Lead Developer, 2025 – Present*

- Vehicle auction and marketplace platform built as a pnpm monorepo (frontend, Hono API, Supabase backend, crawler); leading end-to-end build as the contracted developer (**150+ merged PRs**).

Technical Skills

Research Methods: Dynamic Program Analysis, Mining Software Repositories (SZZ), Statistical Evaluation (Kendall τ, Cohen's κ, PABAK), Inter-rater Agreement

Languages: TypeScript, JavaScript, Python, Java, C#, C++

Web & Frontend: React 18/19, Next.js, Angular, Node.js, Hono, Express, FastAPI, Spring Boot 3, ASP.NET Core

Testing & Automation: Playwright, axe-core, Vitest, Karma, Mocha, Chai, Sinon, Selenium

ML / NLP: TensorFlow, HuggingFace Transformers, BPE Tokenization (own impl.), TextRank, TF-IDF, scikit-learn, pandas

DevOps / Cloud: AWS CodeArtifact, AWS S3, Azure, Cloudflare Workers, Docker, NGINX, GitHub Actions

Databases: MongoDB, PostgreSQL (Supabase), SQL Server, Firebase, Redis

Awards & Achievements

ICT Division Fellowship, Government of Bangladesh: research funding for M.Sc. thesis (Grant No. 56.00.0000.052.33.002.24-61, Apr 2026)

Winner, Optimizely Global Hackathon 2025 (company-wide international hackathon; team built Spotlight, shipped to CMP)

Top 6, ICSE SCORE 2023 (international undergraduate SE competition)